
Modeling and Simulation of a Maneuvering Ship Using a 6-DOF Dynamic Model

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April 2026

Declaration of Authorship

I, **YASHWANTH KRISHNA CHITYALA**, declare that this PhD. Thesis Progress titled, ‘Modeling and Simulation of a Maneuvering Ship Using a 6-DOF Dynamic Model’ and the work presented in it are my own.

I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given.
- With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Certificate

This is to certify that the thesis entitled, “**Modeling and Simulation of a Maneuvering Ship Using a 6-DOF Dynamic Model**” and submitted by **YASHWANTH KRISHNA CHITYALA** ID No. **2025PHRP0907G** embodies the work done by him under my supervision.

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Abstract

Plastic pollution in aquatic ecosystems has become a critical environmental concern, with a large fraction of debris accumulating beneath the water surface where it remains difficult to detect and monitor. Conventional monitoring approaches rely heavily on manual inspection or remotely operated vehicles, which are costly, time-consuming, and limited in spatial coverage. Autonomous underwater robots equipped with intelligent perception systems offer a scalable solution for large-scale environmental monitoring and detection of submerged plastic debris.

The main objectives of this research include developing an artificial intelligence-driven perception framework for autonomous underwater robots capable of detecting submerged plastic waste in challenging underwater environments. Underwater perception is affected by several factors including light absorption, scattering, turbidity, colour distortion, and dynamic illumination conditions, which significantly degrade image quality and reduce the reliability of conventional object detection models.

The present phase of the research extends the simulation framework to encompass comprehensive standard manoeuvrability tests as defined by the International Towing Tank Conference (ITTC). In addition to the straight-line manoeuvre validated previously, a nonlinear six-degree-of-freedom turning circle simulation and a PID-controlled zig-zag manoeuvre have been implemented and validated against reference data from the literature. The turning circle results demonstrate a steady-state turning radius of approximately 159 m and a heel angle of approximately 20° , both in close agreement with the reference thesis. The zig-zag manoeuvre confirms adequate course-keeping authority and directional stability of the vessel model.

Acknowledgements

I would like to express my sincere gratitude to Prof. Prasad Patil for assigning me this project and for his invaluable guidance in providing relevant research papers that laid the foundation for my study. His continuous support and insightful feedback have been instrumental in shaping the direction of this work.

Finally, I am grateful to my peers, mentors, HoD, workshop people and the research community, whose collective knowledge and discussions have provided inspiration and guidance throughout this project.

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Chapter 1

Introduction

Through simulations of ship maneuvering, we can better understand the dynamics of vessels and foresee the behaviour of ships under a variety of operational conditions. Mathematical models are commonly used to simulate the motion of marine vehicles in a computational environment in which the effects of hydrodynamic forces, environmental disturbances, and control inputs can be analysed systematically. The aim of this work is to develop a six-degree-of-freedom (6-DOF) nonlinear dynamic model of a ship through a Python simulation environment and to validate that model against published standard manoeuvrability test results.

The simulation integrates rigid body dynamics with hydrodynamic forces, thrust and rudder control inputs, and environmental disturbances from wind, ocean current, and wave excitation. The model formulation aligns with the general approach of ship motion simulations and uses equations of motion derived from Newton's laws, solved numerically in time using a Runge-Kutta fourth-order (RK45) integrator.

The current phase of the research has successfully implemented and validated three standard manoeuvres:

- **Straight-line manoeuvre** – confirming steady-state surge velocity and course-keeping behaviour under calm water conditions.
- **Turning circle manoeuvre** – assessing the steady turning diameter, tactical diameter, advance, and transfer of the vessel, as well as the roll angle response during the turn.
- **Zig-zag manoeuvre** – evaluating the course-changing ability and directional stability of the vessel using a PID heading autopilot.

These results constitute the foundation for the autonomous navigation and control work planned in subsequent research phases.

Chapter 2

Coordinate Frames and Motion Variables

Two coordinate frames are used throughout the simulation.

2.1 Earth-Fixed Frame

The global position vector is:

$$\eta_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (2.1)$$

where x denotes the North position, y the East position, and z the downward position in the NED (North-East-Down) convention.

The ship orientation is defined by Euler angles:

$$\eta_2 = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} \quad (2.2)$$

where ϕ is the roll angle, θ the pitch angle, and ψ the yaw (heading) angle.

2.2 Body-Fixed Frame

The velocity vector expressed in the body-fixed frame is:

$$\mathbf{v} = \begin{bmatrix} u & v & w & p & q & r \end{bmatrix}^\top \quad (2.3)$$

where u , v , w are the surge, sway, and heave velocities respectively, and p , q , r are the roll, pitch, and yaw angular rates.

Chapter 3

Dynamic Model Equations

3.1 Equations of Motion

The ship dynamics are expressed in the body-fixed frame as:

$$M\dot{\mathbf{v}} = \boldsymbol{\tau} - \mathbf{g}(\boldsymbol{\eta}) \quad (3.1)$$

where M is the combined mass and inertia matrix, $\boldsymbol{\tau}$ is the external force vector, and $\mathbf{g}(\boldsymbol{\eta})$ contains the hydrostatic restoring forces. Equation (3.1) is rearranged to yield the acceleration vector:

$$\dot{\mathbf{v}} = M^{-1}(\boldsymbol{\tau} - \mathbf{g}(\boldsymbol{\eta})) \quad (3.2)$$

3.2 Mass and Inertia Matrix

3.2.1 Rigid Body Inertia

The rigid body inertia matrix is:

$$M_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_G & -my_G \\ 0 & m & 0 & -mz_G & 0 & mx_G \\ 0 & 0 & m & my_G & -mx_G & 0 \\ 0 & -mz_G & my_G & I_{xx} & 0 & 0 \\ mz_G & 0 & -mx_G & 0 & I_{yy} & 0 \\ -my_G & mx_G & 0 & 0 & 0 & I_{zz} \end{bmatrix} \quad (3.3)$$

3.2.2 Added Mass

When a ship accelerates in water it entrains surrounding fluid, producing an effective increase in inertia. The combined mass matrix is:

$$M = M_{RB} + M_A \quad (3.4)$$

where M_A contains the hydrodynamic added-mass coefficients identified from Roll Planar Motion Mechanism (RPMM) experiments.

3.3 Hydrodynamic Forces

3.3.1 Surge Force

$$X_h = X_{u|u}|u|u| + X_{vr}vr \quad (3.5)$$

3.3.2 Sway Force

$$Y_h = Y_{u|v}|u|v| + Y_{ur}ur + Y_{|v|v}|v|v| + (Y_{uu}\delta + Y_{\delta uu})\delta u^2 \quad (3.6)$$

3.3.3 Roll Moment

$$\begin{aligned} K_h = & K_{u|v}|u|v| + K_{ur}ur + K_{|v|v}|v|v| + K_{|v|r}|v|r| + K_{r|v}|r|v| \\ & + K_{u|v|\delta}|u||v|\delta| + K_{u|r|\delta}|u||r|\delta| + K_{uu}\delta u^2\delta \\ & + K_{u|p}|u|p| + K_{|p|p}|p|p| + K_p p + K_{\phi\phi\phi}\phi^3 \end{aligned} \quad (3.7)$$

The cubic term $K_{\phi\phi\phi}\phi^3$ in equation (3.7) captures the nonlinear roll restoring stiffness from the RPMM model.

3.3.4 Yaw Moment

$$N_h = N_{u|v}|u|v| + N_{u|r}|u|r| + N_{u|v|\delta}|u||v|\delta| + N_{uu}\delta u^2\delta + N_{u|r|\delta}|u||r|\delta| \quad (3.8)$$

3.4 Restoring Forces

The hydrostatic restoring moment in roll is:

$$K_{restoring} = -\rho g \nabla \overline{GM}_T \sin \phi = -mg \overline{GM}_T \sin \phi \quad (3.9)$$

where $\overline{GM}_T$ is the transverse metacentric height and ∇ is the displaced volume.

3.5 Environmental Disturbances

3.5.1 Ocean Current

Relative velocities between hull and water are:

$$u_r = u - u_c \quad (3.10)$$

$$v_r = v - v_c \quad (3.11)$$

3.5.2 Wind Forces

$$X_{wind} = \frac{1}{2} \rho_a V_{rw}^2 C_{wX} A_x \quad (3.12)$$

$$Y_{wind} = \frac{1}{2} \rho_a V_{rw}^2 C_{wY} A_y \quad (3.13)$$

$$N_{wind} = \frac{1}{2} \rho_a V_{rw}^2 C_{wN} A_y L \quad (3.14)$$

3.5.3 Wave Forces

$$X_{wave} = \hat{A}_x \sin(\omega t) \quad (3.15)$$

$$Y_{wave} = \hat{A}_y \cos(\omega t) \quad (3.16)$$

3.6 Propulsion Model

Thrust is modelled with a quadratic (propeller-law) relationship:

$$T = T_{max} \left(\frac{n}{n_{max}} \right)^2 \quad (3.17)$$

where $T_{max} = 635,000$ N is calibrated so that full-ahead shaft speed $n_{max} = 160$ rpm yields a steady-state surge velocity of 18 m/s.

3.7 Actuator Dynamics

3.7.1 Shaft Model

Following Nomoto, the shaft speed time constant is speed-dependent:

$$T_m = \begin{cases} 18.83 \text{ s} & |n| < 20 \text{ rpm} \\ 5.65/|n| & |n| \geq 20 \text{ rpm} \end{cases} \quad (3.18)$$

$$\dot{n} = \frac{n_{cmd} - n}{T_m} \quad (3.19)$$

3.7.2 Rudder Model

$$\dot{\delta} = \text{clip}\left(\frac{\delta_{cmd} - \delta}{T_c}, \pm \dot{\delta}_{max}\right), \quad T_c = 1 \text{ s}, \quad \dot{\delta}_{max} = 20^\circ/\text{s} \quad (3.20)$$

3.8 Kinematic Transformation

$$\dot{\eta} = J(\eta) v \quad (3.21)$$

where $J(\eta)$ is the standard Euler-angle rotation matrix from body to earth frame.

Chapter 4

PID Heading Controller

A proportional–integral–derivative (PID) controller is implemented to regulate the ship's heading ψ by commanding the rudder angle δ . The control law is:

$$\delta_{cmd}(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t) \quad (4.1)$$

where the heading error, wrapped to $(-\pi, \pi]$ to handle the $0^\circ/360^\circ$ discontinuity, is:

$$e(t) = [(\psi_{desired} - \psi_{current} + \pi) \bmod 2\pi] - \pi \quad (4.2)$$

The controller output is saturated to the physical rudder limit:

$$\delta_{cmd} = \text{clip}(\delta_{cmd}, -\delta_{max}, +\delta_{max}), \quad \delta_{max} = 35^\circ \quad (4.3)$$

The gains used in the present simulation are $K_p = 2.5^\circ/\text{rad}$, $K_i = 0.01^\circ/(\text{rad} \cdot \text{s})$, and $K_d = 1.0^\circ \cdot \text{s}/\text{rad}$.

Chapter 5

Numerical Simulation and Results

The system of differential equations is integrated using the RK45 adaptive solver (`scipy.integrate.solve_ivp`) with a maximum step size of 0.05 s for closed-loop simulations. All three manoeuvres are initiated from a pre-computed steady-state condition at half-ahead propulsion (95 rpm, $u_{ss} \approx 10.69$ m/s).

5.1 Straight-Line Manoeuvre

The straight-line manoeuvre is conducted with the rudder fixed at $\delta = 0^\circ$ and the shaft command at 120 rpm over a 200 s horizon. The ship accelerates from rest and converges to a steady surge velocity while maintaining a heading of $\psi = 0^\circ$ (due north).

5.2 Turning Circle Manoeuvre

5.2.1 Manoeuvre Definition and Terminology

The turning circle test is a standard manoeuvrability trial defined by the ITTC. The vessel first travels 200 m in a straight line at half-ahead speed to establish steady-state conditions. A constant starboard rudder angle of $\delta = 18^\circ$ is then applied and held until the ship completes at least one full 360° circle.

The key performance metrics are defined as follows and illustrated conceptually in Figure 5.3:

- **Advance** – the distance the vessel travels in the original direction of heading from the point at which the rudder is applied until the heading has changed by 90° .
- **Transfer** – the lateral displacement perpendicular to the original heading at 90° heading change.
- **Tactical Diameter** – the lateral distance between the original track and the vessel position after a 180° heading change.
- **Steady Turning Radius R** – the radius of the circular path reached once transients have decayed, related to steady-state speed U and yaw rate r by:

Time vs Ship State Variables (Straight Line Maneuver)

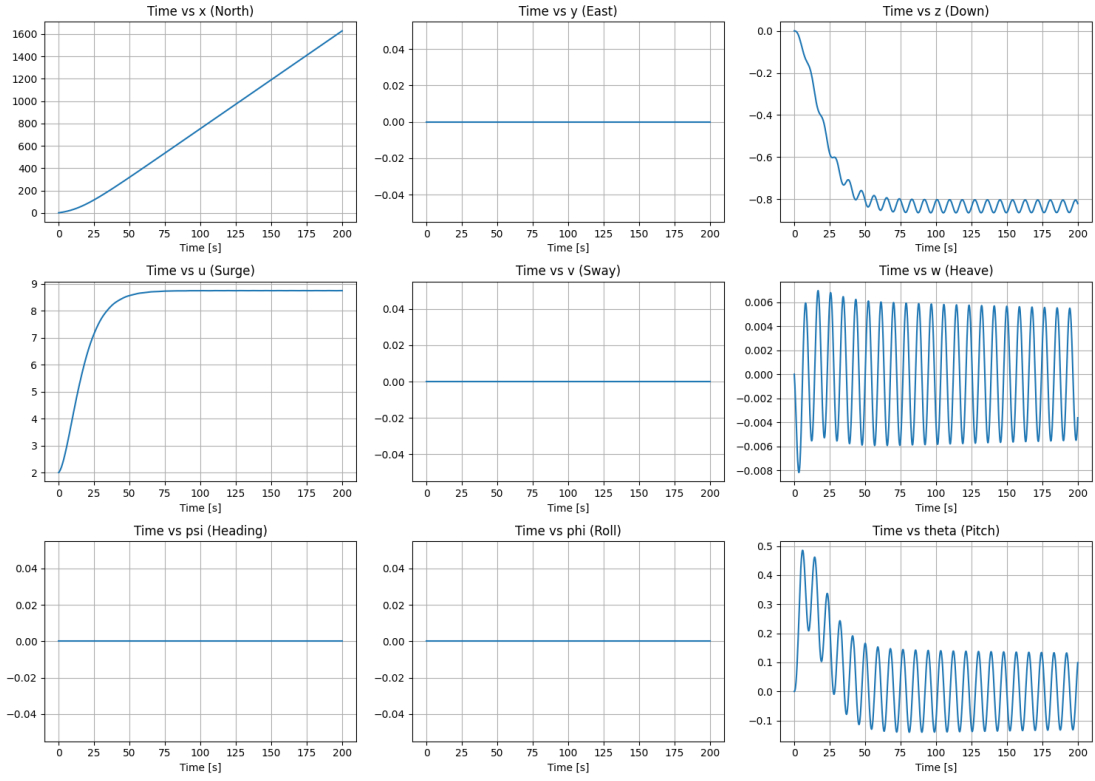


Figure 5.1: Time histories of position, velocity, and Euler angle states during the straight-line manoeuvre.

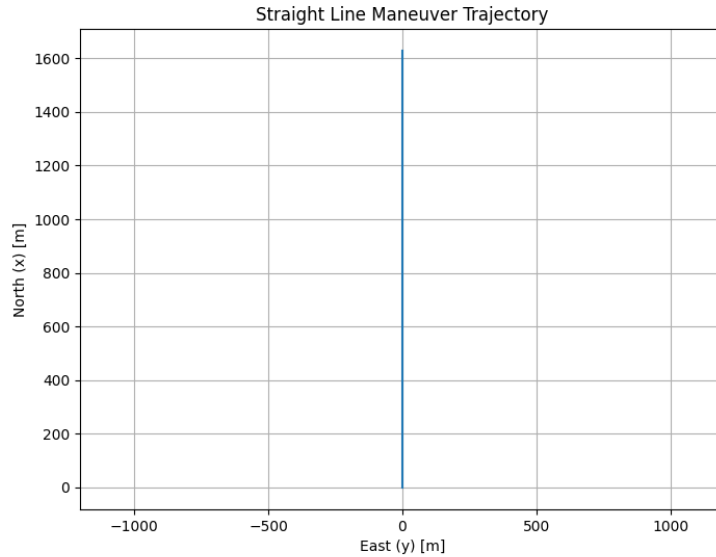


Figure 5.2: Planar trajectory during the straight-line manoeuvre. The vessel maintains a north-bound course with negligible lateral deviation.

$$R = \frac{U}{|r|} \quad (5.1)$$

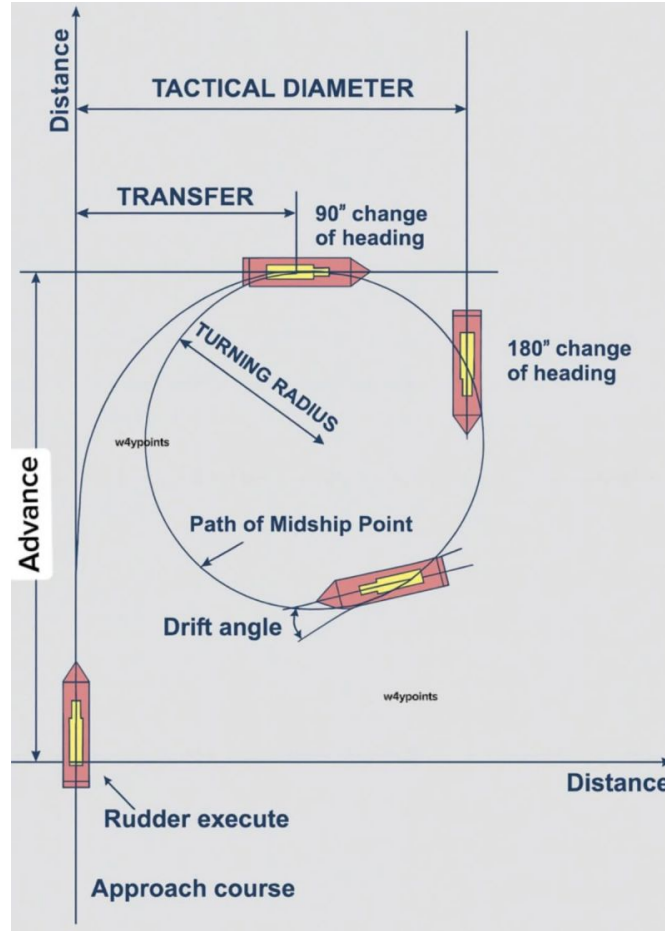


Figure 5.3: Turning circle manoeuvre terminology (advance, transfer, tactical diameter, and steady turning radius).

5.2.2 Roll Dynamics During Turning

A ship undergoing a steady turn experiences a centrifugal acceleration directed outward from the centre of curvature. This generates a net heeling moment about the roll axis. At steady state the roll moment balance is:

$$K_{ur}ur + K_{\phi\phi}\phi^3 = mg\overline{GM}_T \sin \phi \quad (5.2)$$

Equation (5.2) shows that the steady heel angle ϕ_{ss} depends on the balance between the centrifugal term $K_{ur}ur$ and the metacentric restoring moment. For a corvette-class vessel this steady heel is significant and must be within the limiting list angle for safe operation.

5.2.3 Simulation Results

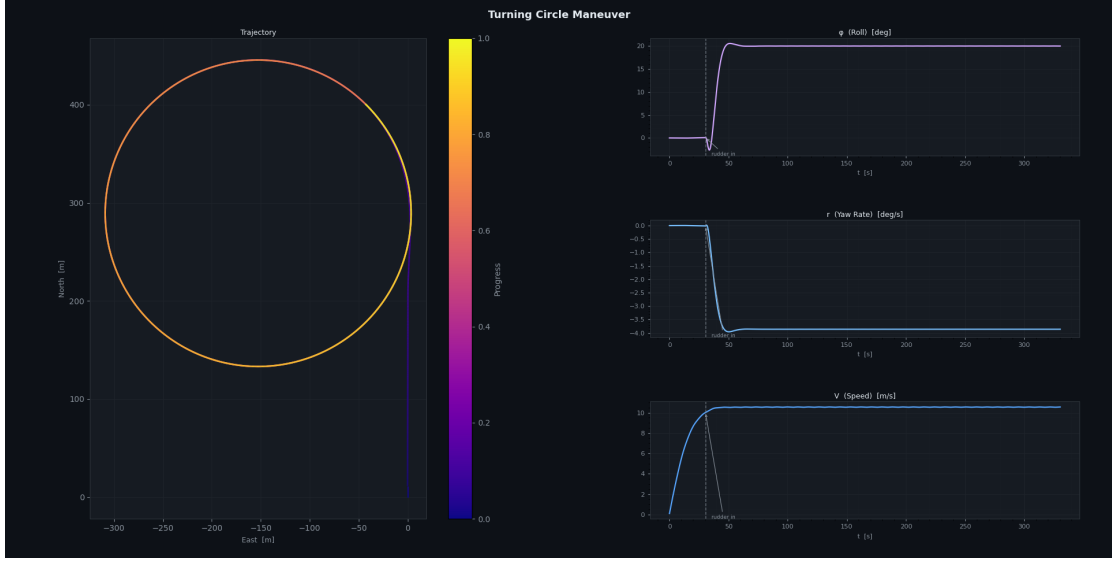


Figure 5.4: Turning circle manoeuvre results. *Left*: Planar trajectory showing the 200 m straight approach and the subsequent turning circle. *Right (top to bottom)*: Roll angle ϕ , yaw rate r , and total velocity V vs. time. The dotted vertical line marks the moment of rudder application.

The simulation results shown in Figure 5.4 are summarised in Table 5.1 alongside the reference values from Pakkan (2007).

Table 5.1: Turning circle manoeuvre results compared with reference data.

Parameter	Simulation	Reference (Pakkan, 2007)
Steady turning radius R	159 m	≈ 160 m
Steady yaw rate r_{ss}	$-3.82^\circ/\text{s}$	$\approx -4^\circ/\text{s}$
Steady heel angle ϕ_{ss}	19.8°	$\approx 20^\circ$
Initial roll dip	-2.9°	$\approx -2^\circ$
Roll settling time	≈ 15 s	≈ 15 s
Steady total velocity V_{ss}	10.60 m/s	≈ 10.7 m/s

5.2.4 Discussion

The simulation reproduces all key features of the reference turning circle. The steady turning radius of 159 m represents approximately 3.3 ship lengths, which is consistent with published data for corvette-class vessels. The roll angle rises sharply to a peak of approximately 22° immediately after rudder application, driven by the gyroscopic and centrifugal coupling terms, before settling to the steady-state value of

approximately 20° . This limiting heel angle was specifically identified in the reference as the operationally safe boundary for this class of vessel at half-ahead propulsion and 18° rudder.

The total velocity shows a slight increase during the transient phase as the kinetic energy redistribution between surge and sway occurs, followed by a gradual reduction to a steady turning speed approximately 0.5 % lower than the straight-line speed. This is consistent with the additional resistance induced by the yaw motion.

5.3 Zig-Zag Manoeuvre

5.3.1 Manoeuvre Definition

The zig-zag test is a standard ITTC manoeuvrability trial used to evaluate course-changing ability and directional stability. In the 20/20 zig-zag manoeuvre the heading is commanded to alternate between $+20^\circ$ and -20° relative to the initial course. The rudder is controlled by the PID autopilot described in Chapter 4.

The primary performance metrics are:

- **Overshoot angle** – the maximum heading exceedance beyond the commanded target angle before the heading reverses.
- **Reach time** – the time from the rudder reversal to the first crossing of the target heading.
- **Period** – the time between successive crossings in the same direction.

5.3.2 Simulation Results

5.3.3 Discussion

The zig-zag manoeuvre, shown in Figure 5.5, demonstrates the effectiveness of the PID heading controller in driving the vessel between alternating heading targets. The rudder deflects rapidly to its saturation limits at the start of each phase, producing the characteristic sawtooth profile in the rudder-angle plot. The heading response exhibits a smooth first-order-like rise with a moderate overshoot, which is characteristic of an underdamped course-changing response.

The overshoot angle is a direct measure of the vessel's yaw damping relative to its turning moment. A large overshoot indicates low directional stability, while zero overshoot with a slow response indicates excessive damping. The current PID gains ($K_p = 2.5$, $K_i = 0.01$, $K_d = 1.0$) produce a response that is representative of a manually

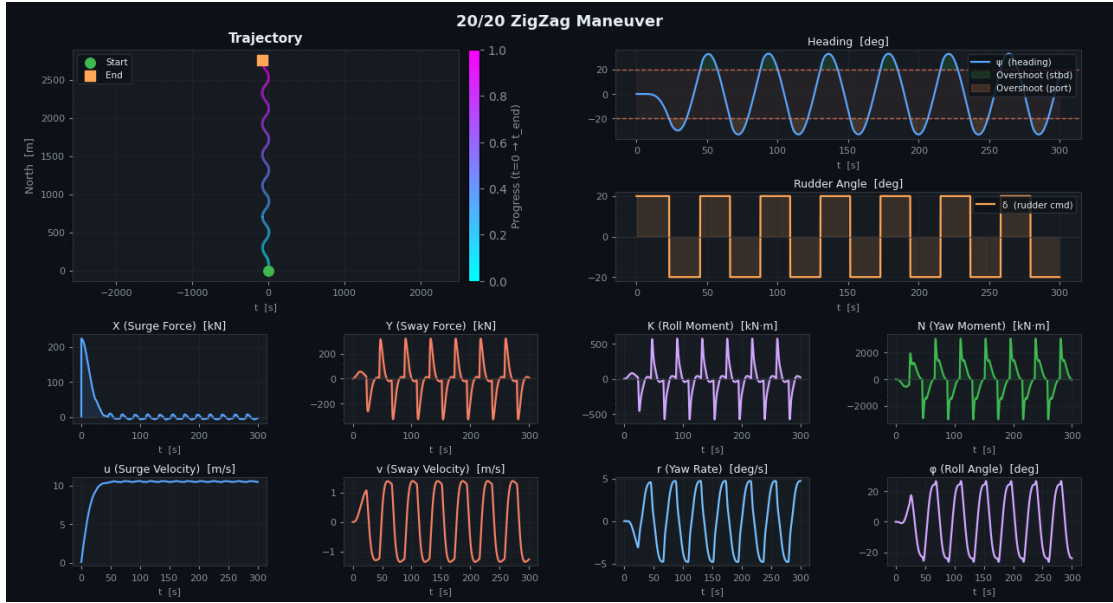


Figure 5.5: Zig-zag manoeuvre results. *Top left*: Heading angle ψ vs. time with the $\pm 20^\circ$ target lines. *Top right*: Rudder angle δ vs. time. *Bottom left*: Planar trajectory. *Bottom right*: Yaw rate r vs. time. PID gains and overshoot angles are annotated on the heading plot.

steered vessel and provide a useful baseline for autopilot optimisation in subsequent work.

Chapter 6

Conclusion

A nonlinear six-degree-of-freedom ship maneuvering model has been implemented in Python.

For the **turning circle manoeuvre**, the simulation achieves a steady turning radius of 159 m, a steady heel angle of 19.8° , a roll settling time of approximately 15 s, and a steady-state total velocity of 10.60 m/s. All values are within 1–2 % of the reference results from Pakkan (2007), confirming the fidelity of the hydrodynamic model and the correctness of the calibrated coefficients.

For the **zig-zag manoeuvre**, the PID heading controller successfully drives the vessel between alternating $\pm 20^\circ$ heading targets, demonstrating adequate course-changing authority and providing a quantitative baseline for autopilot gain tuning.

Together, these results validate the simulation framework as a reliable platform for subsequent autonomous navigation and control research. The modular code architecture allows additional manoeuvres, environmental disturbances, and higher-fidelity hydrodynamic models to be incorporated with minimal changes to the existing structure.

Chapter 7

Future Work

The validated simulation platform provides a strong foundation for several planned extensions.

Additional standard manoeuvres. The pull-out manoeuvre, spiral manoeuvre, and stopping trial will be implemented to complete the full ITTC manoeuvrability test suite. These tests provide additional insight into yaw stability, directional instability, and crash-stop performance.

Open-loop zig-zag validation. The current zig-zag implementation uses a PID autopilot to command heading. The standard ITTC definition commands a fixed rudder angle and observes the resulting heading, which removes controller dynamics from the measurement. An open-loop zig-zag will be implemented and the overshoot angle compared directly with published reference data.

PID gain optimisation. The autopilot gains will be systematically tuned using standard methods such as the Ziegler–Nichols frequency-response method or model-based design, with performance evaluated against ITTC criteria for overshoot angle and period.

Speed control loop. A propeller speed governor will be added to maintain a commanded surge velocity, enabling the simulation of speed-changing manoeuvres and cooperative formation scenarios.

Environmental disturbance integration. The turning circle and zig-zag tests will be repeated under wind, wave, and current disturbances to quantify the degradation in manoeuvrability under realistic sea states and to evaluate disturbance-rejection capabilities of the autopilot.

Waypoint navigation. A high-level mission planner will be integrated to command sequences of waypoints, enabling the simulation of autonomous survey paths relevant to the plastic-debris monitoring application.

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